

ESPark

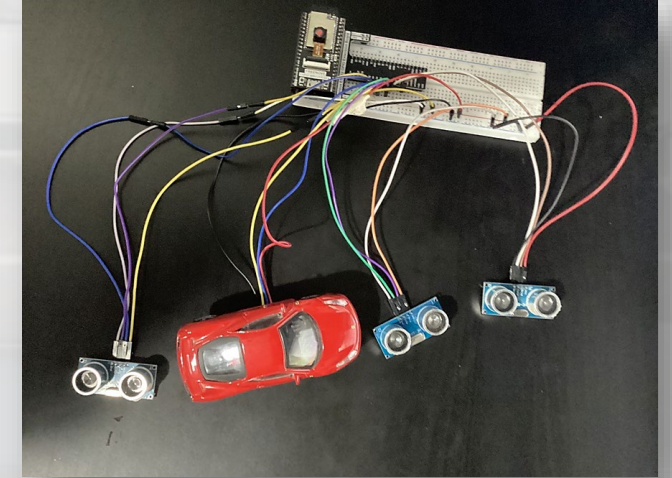


Lorenzo Russo - 2091186

Introduction

A SIMULATION OF THE PARKING LOTS WITH AN ESP32 MICROCONTROLLER

The main goal of the app is to make users avoid endless searches for a parking space; by using our app the user can first view which parking space is free, in particular he/she will be able to locate the free parking space in certain sectors or floors of the parking lot, by doing so he/she will be able to reach the available parking slot optimizing the search.



The background of the slide is a photograph of a parking lot, split diagonally from the top-left to the bottom-right. The upper-left portion shows an indoor parking garage with several cars parked in rows. The lower-right portion shows an outdoor parking lot with many cars parked in rows, with a yellow pedestrian crossing sign visible on the right. A large, light gray arrow points downwards from the center of the diagonal split towards the word 'COMPETITORS'.

COMPETITORS

Competitors



EasyPark



Parkopedia



Telepass



EasyPark makes it possible to reserve a parking space in advance so that the user, once he arrives in the car park and has chosen his own parking space, can park.

Pro:

The user can pay for parking directly from his smartphone.

Using the FIND function, it allows you to view the percentage of occupancy of the parking lots.

Cons:

It forces the user to insert a credit card even for free parking.

communicates the percentage of parking spaces occupied only

with respect to vehicles whose drivers use the app.

We can post reviews, book or otherwise help grow the community of the service

Pro:

A lot of filters such as the duration of the stay, the payment methods, the presence of charging stations for electric cars and so on.

Cons:

The amount of options is not a real advantage as to use it you need to have a premium account

Telepass application to search for a parking space directly from the application

Pro:

With the device you will be able to enter and exit the car parks in the structure in the reserved lane without buying a ticket

Cons:

To use the service you must choose only the parking space with telepass logo, only certified parking space, and not just a free one.

Competitors

	ESPark	EasyPark	Parkopedia	Telepass
Account (profile, vehicles owned, payment methods)	✓	✓	✗	✓
Map to choose the parking	✓	✓	✓	✓
Filters to choose the suitable car park	✓	✗	✓	✗
Pay for parking with the app	✓	✓	✗	✓
Extend your parking duration with the app	✓	✓	✗	✓
Real-time update of the availability or non-availability of a parking slot	✓	✗ (only a percentage estimate of the availability of the entire car park)	✗	✗



USER PROFILE, PERSONAS AND SCENARIO



User Profile



Age	14-65 years
Gender	Any
Occupation	Any
Location	Any
Technology	Smartphone with GPS and internet connection
Vehicle	Car/motorcycle



Persona 1: Giulia

Giulia is a 29 years old commercial advisor born and grown in Roma in Garbatella district. She recently buy a new car to facilitate her travels to work, theatre classes, cinema, and tennis matches with work colleagues. Since using the car, Giulia has been thinking of moving to a smaller city, as the problem of parking in such a large city has become very stressful for all her activities.

Every morning to go to work Giulia spends 30 minutes to look for a free parking space, plus another 30 when she must look for it as close to home as possible.

Scenario

It's Sunday, Giulia decides to go to visit Aunt Carla in via Tuscolana. To get there from Garbatella it will take more or less 20 minutes by car.

She knows very well that it is very difficult to find parking near her aunt's house. She would like to see through her smartphone if there is a free parking lot before leaving her house. She won't have to waste time looking parking and spending more time with the aunt.



Persona 2: Carl

Carl is a 20-year-old Kenyan boy, a chemical engineering student who moved to Rome thanks to a scholarship.

Due to the distance from his home to the university headquarters, he decides to buy a small-engine motorcycle to facilitate his travels.

But still, he's in late for lessons to look for a parking space near the engineering headquarters.

Scenario

It is Monday, lessons start at 9:00 AM and Carl just got out from his house at 8.30. Carl being aware of the traffic in Rome, and of the problems in finding parking. He decides to rely on an application in such a way that once he arrives at the university he has the possibility to view the free parking available closest to him, so as not to miss even a minute of the lesson.



Persona 3: Maria

Maria is 33 years old interior designer from Florence, who recently moved to Rome. She decided not to invest in buying a car given the high costs for the purchase but also for the high management costs.

For the movements she uses public transport, but sometimes she rents a car. Maria, being an interior designer, has a great passion to go to art exhibitions.

Scenario

Maria decides to go with her work colleagues this Sunday to Ikea, but since Ikea is not well connected by public transport, she decides to rent a car. She would like use her smartphone to find an available parking space inside the shopping centre which is free for her arrival.

Her friend has a motor disability, so the application needs to find a parking space that can make it easier for his colleague to get out of the car.



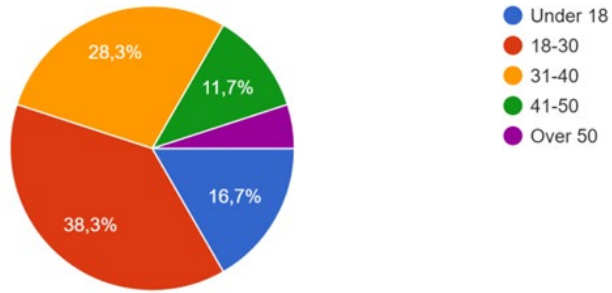
QUESTIONNAIRE ANALYSIS



Questionnaire analysis

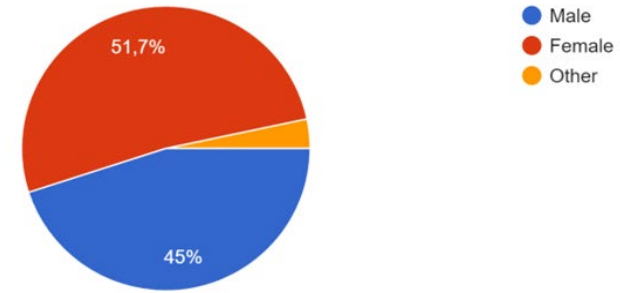
Age

60 risposte



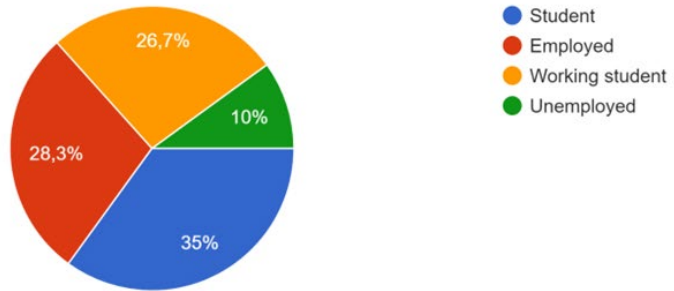
Gender

60 risposte



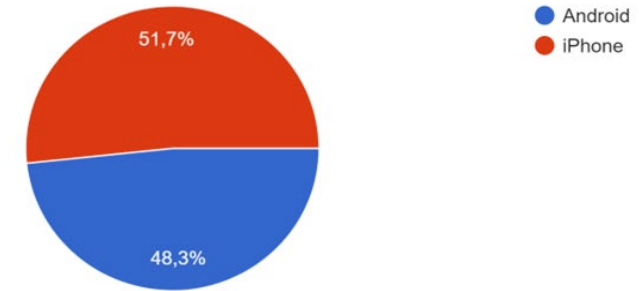
What is your current status?

60 risposte



Type of smartphone?

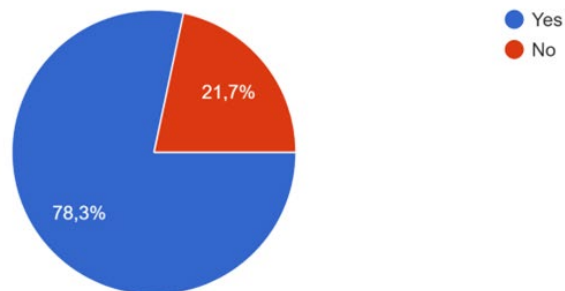
60 risposte



Questionnaire analysis

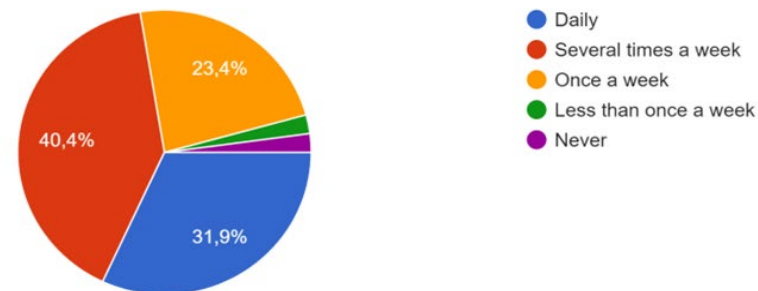
Do you have a driving licence? (of any type, e.g. AM, B, A, ...)

60 risposte



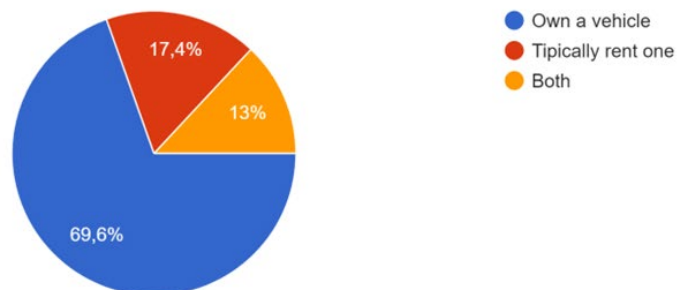
How frequently do you drive to your destination? (ex. to university, to work, to cinema, ...)

47 risposte



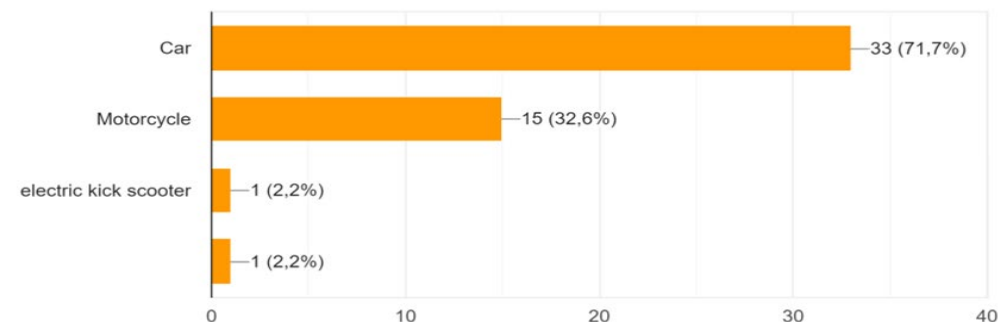
Do you have your own vehicle or do you rent one?

46 risposte



What type of vehicle do you use to reach your destination? (you can select more than one)

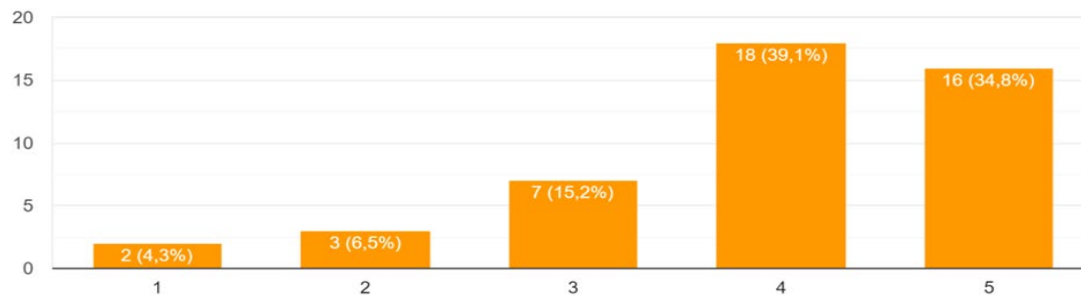
46 risposte



Questionnaire analysis

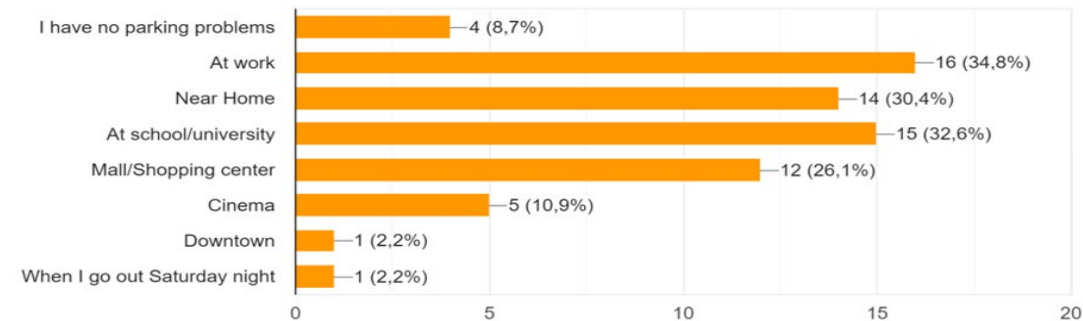
How often do you struggle to find parking in the area you're driving to?

46 risposte



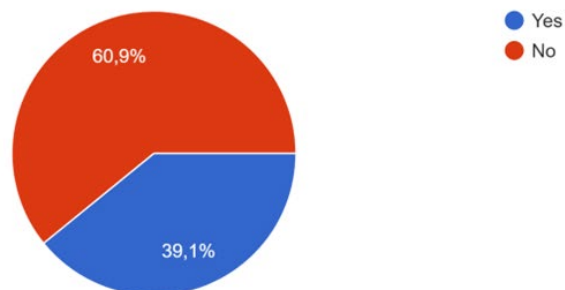
Where do you find it most difficult to find parking? (you can select more than one)

46 risposte



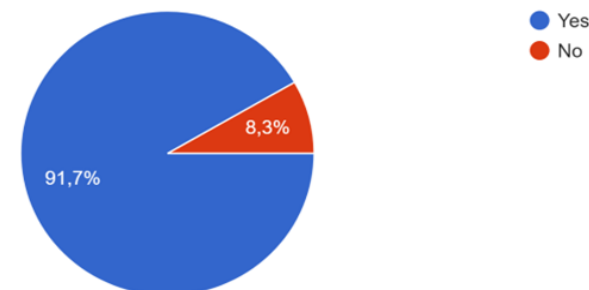
Do you use a parking app?

46 risposte



Do you find an app that shows the availability of parking space in real-time interesting?

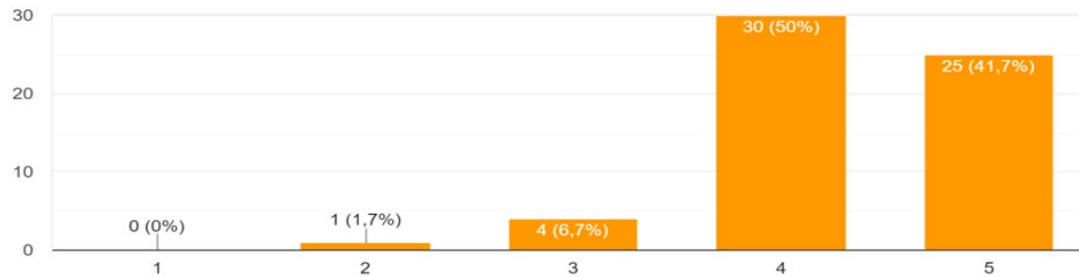
60 risposte



Questionnaire analysis

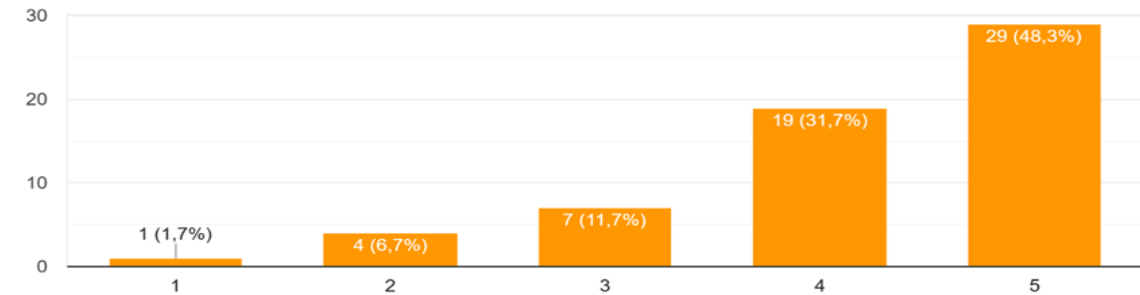
How useful do you think it would be to display the entire parking lot with free and/or occupied spaces, by marking them in red or green?

60 risposte



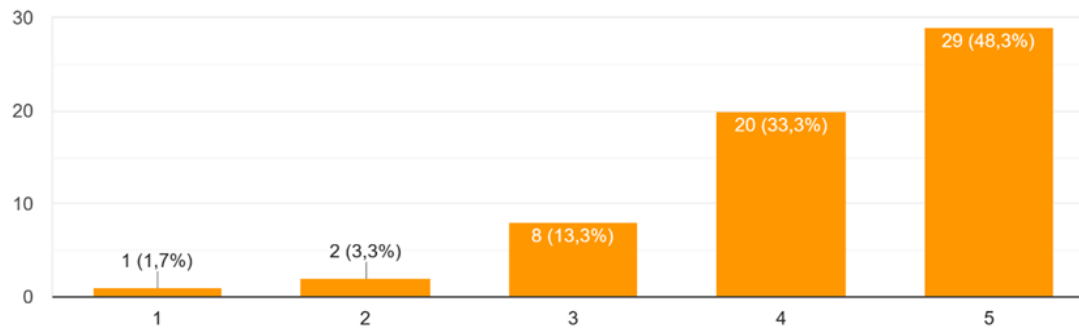
How useful do you think it is for the user to pay for parking directly from the app?

60 risposte



How useful do you think it is for the user to extend the parking duration directly from the app, without returning to the vehicle?

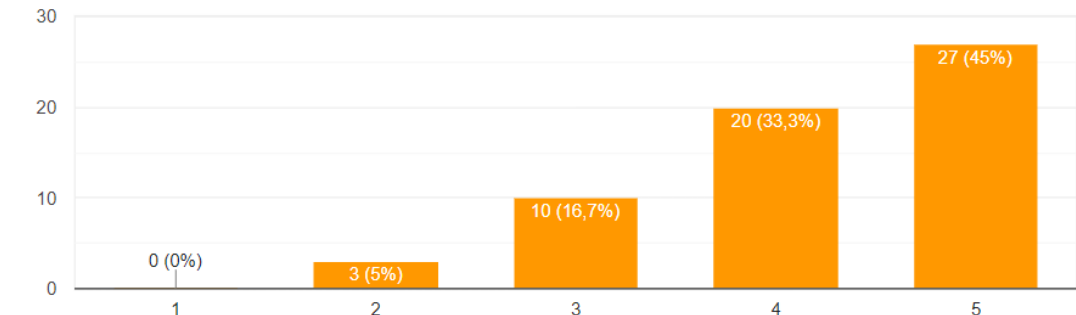
60 risposte



How useful do you think it is for the user to choose an available parking slot before arriving based on need (i.e. closest to the entrance, closest to a shop etc), so that the app can notify the user whether the slot remains available or is occupied before the user arrives?

 Copia

60 risposte

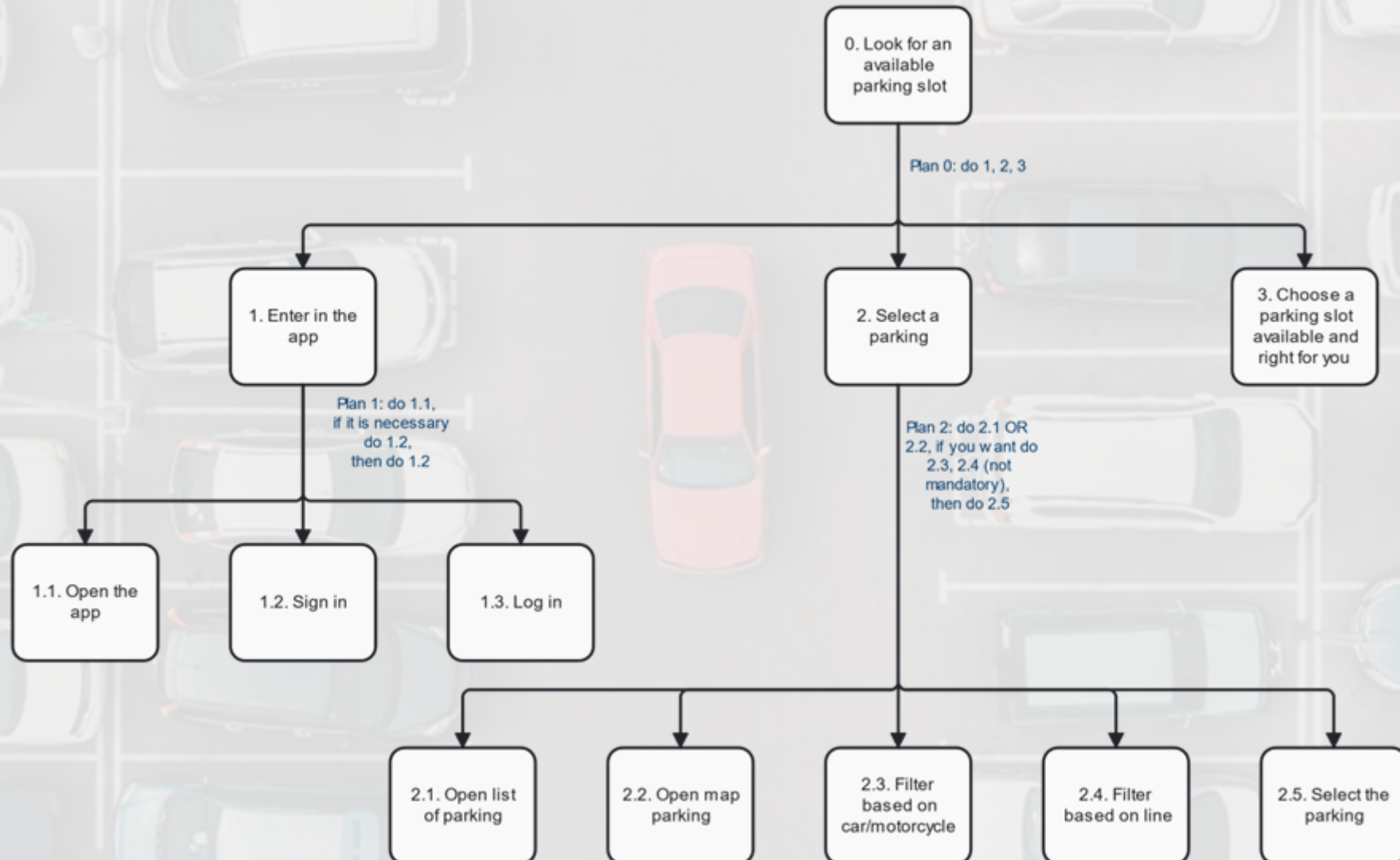


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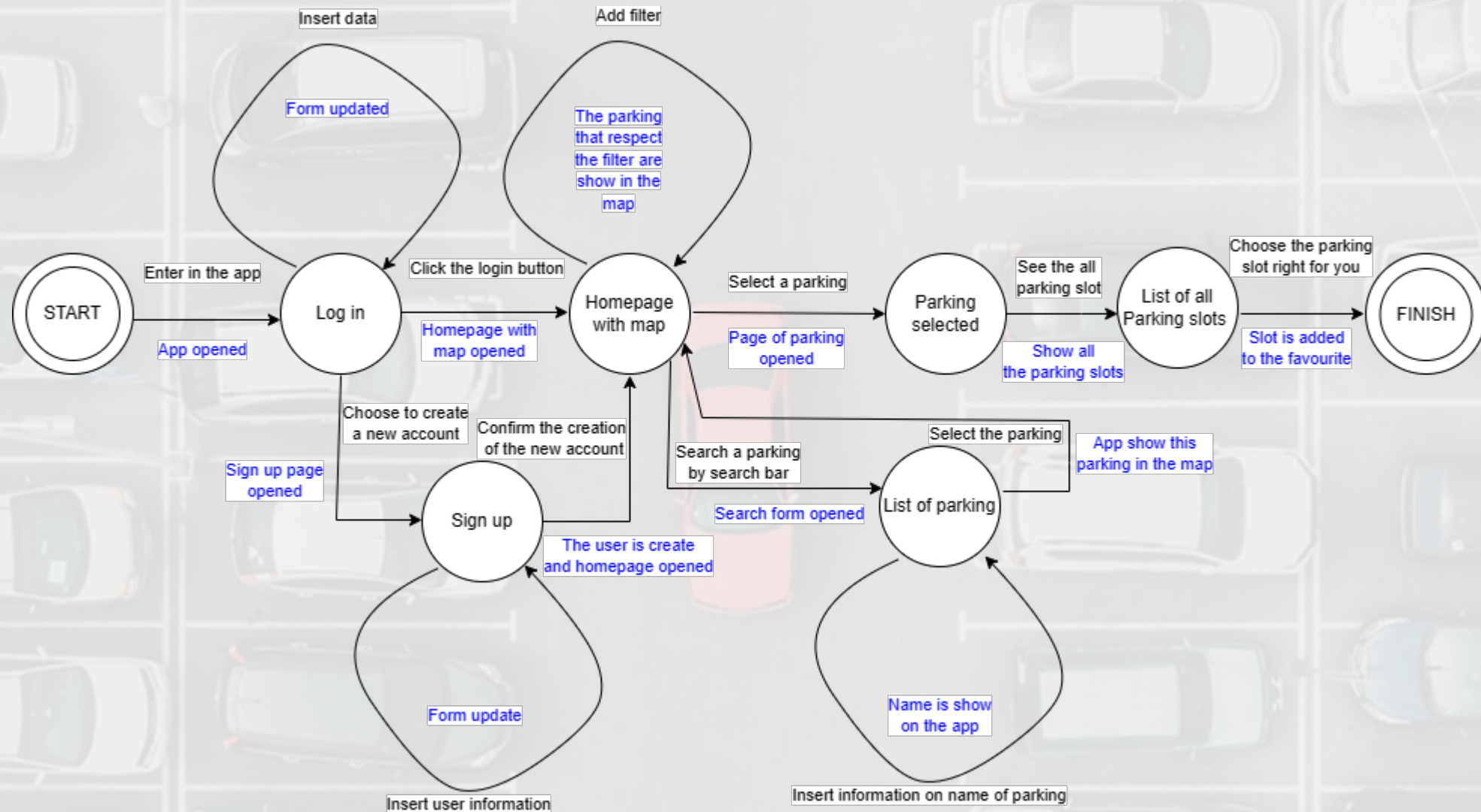
TASK ANALYSIS

HTA: Look for an available parking slot

In order to choose a parking slot the user, after the login, has to select which parking that s/he prefers adding some filtering on the type of vehicle and on colors of lines.

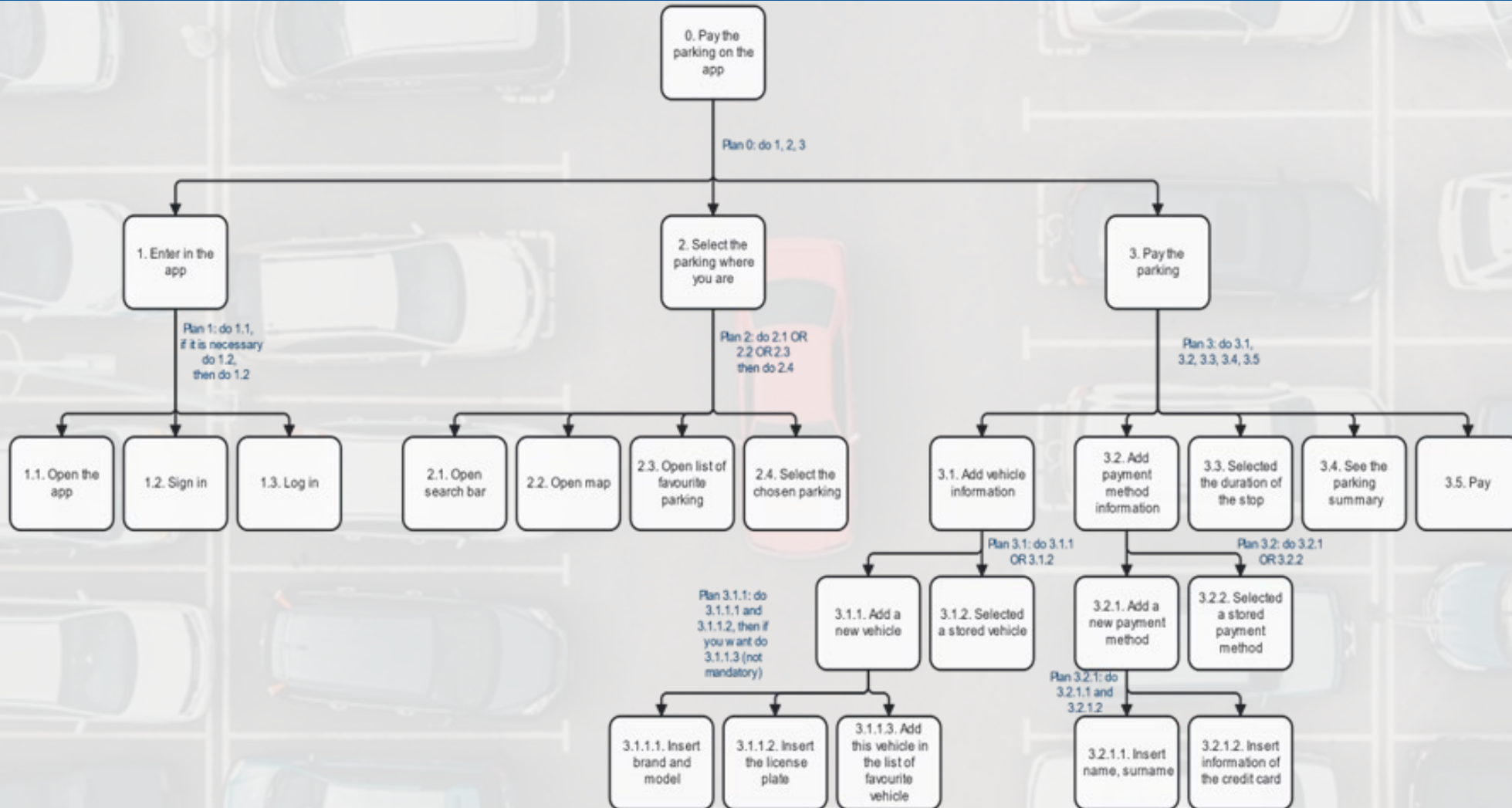


STN: Look for an available parking slot

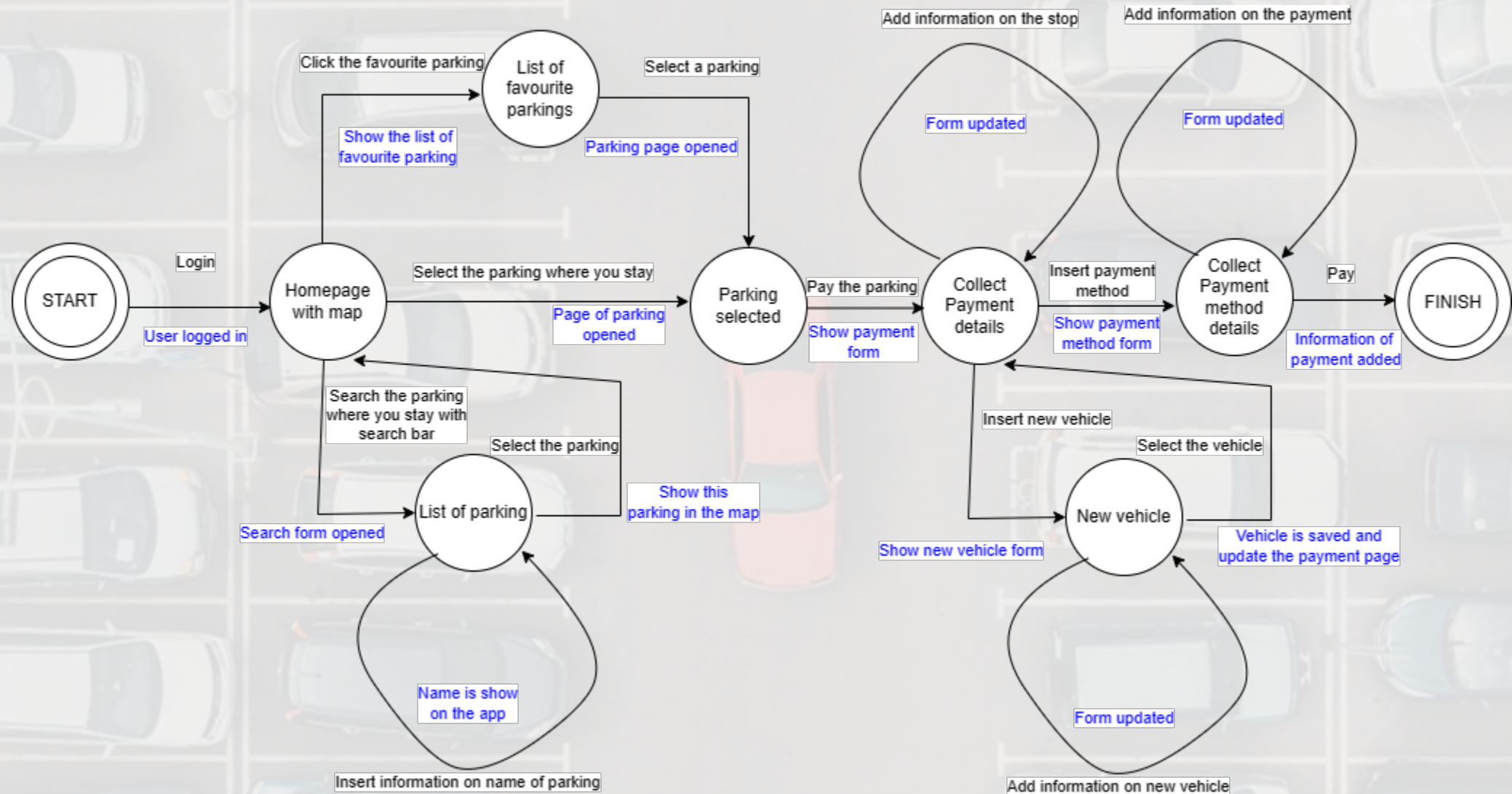


HTA: Pay the parking on the app

After choosing a parking slot the user have to add the driver's license and register the vehicle that he uses, and then can pay with his credit card

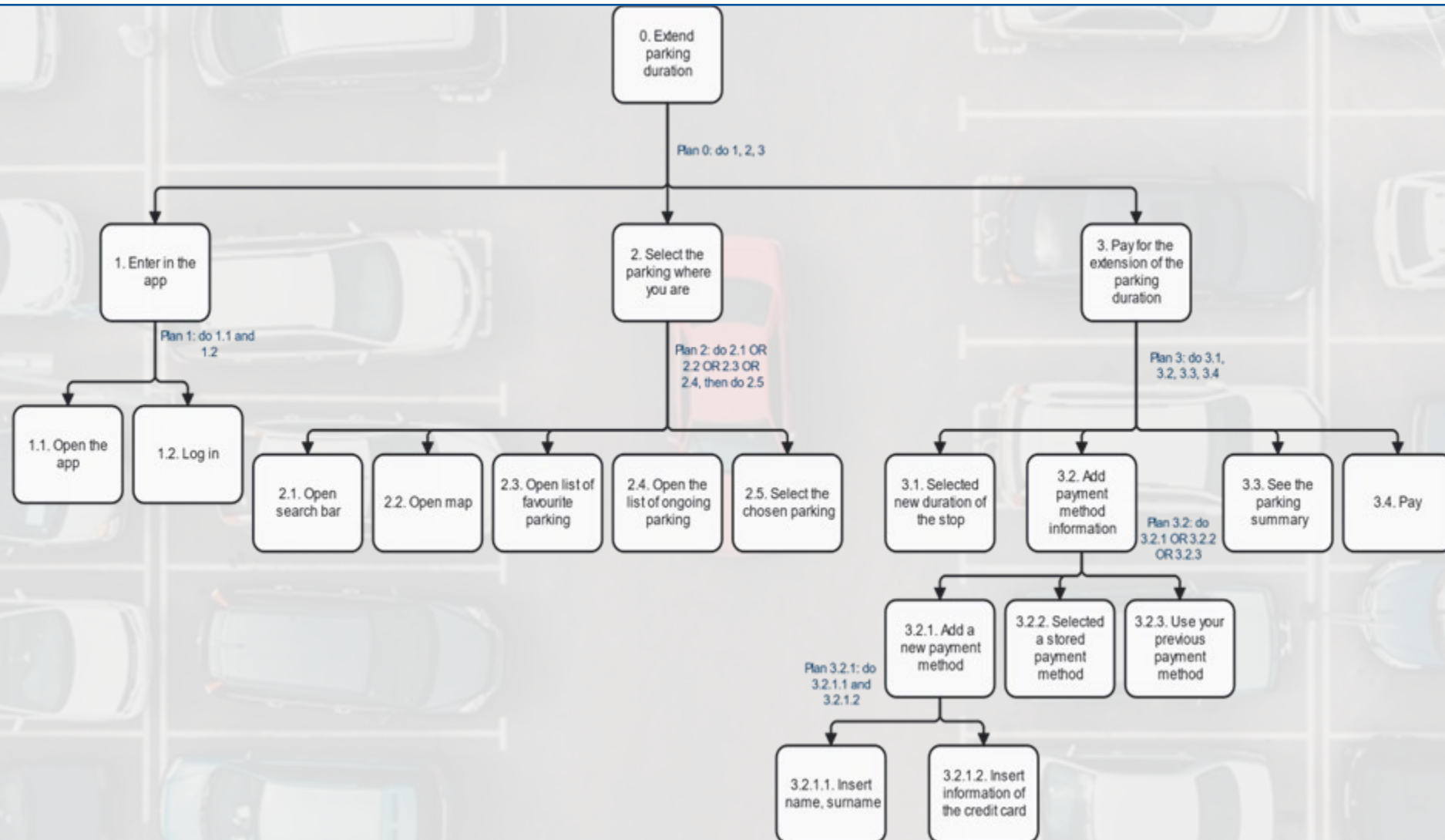


STN: Pay the parking on the app

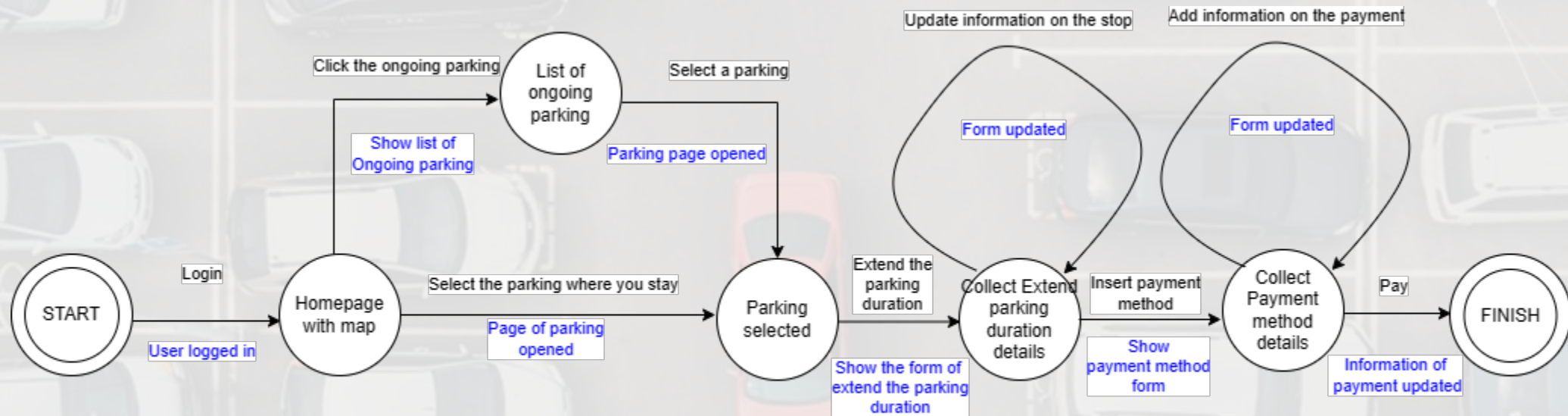


HTA: Extend parking duration

If for every reason the user can't take the car out the deadline time, s/he can extend the duration of stay in the parking slot.



STN: Extend parking duration





PROTOTYPE 1

Prototype 1



Create a new account

Login and view profile

Map, filter, and menù section

Parking page, map of all parking slot

Paying the parking

Extend duration of parking

My parking

Mobile app login screen. The screen displays the "ESPark" logo at the top. Below the logo is a "Login" section with two input fields: "Email" and "Password". A blue "Login" button is positioned below the password field. Below the login button is a link that says "Not account, register". At the bottom of the screen, there are two social media icons (Facebook and Google) and a button that says "Enter without an account".

Mobile app sign-up screen. The screen displays the "ESPark" logo at the top. Below the logo are four input fields: "First Name", "Last Name", "Email", and "Password". Below the password field is a line of text: "By signing up, you agree with the [Terms of Service](#) and [Privacy Policy](#)". Below this text is a button that says "Already have an account?". At the bottom of the screen is a large blue "Sign Up" button.

Prototype 1



Create a new account

Login and view profile

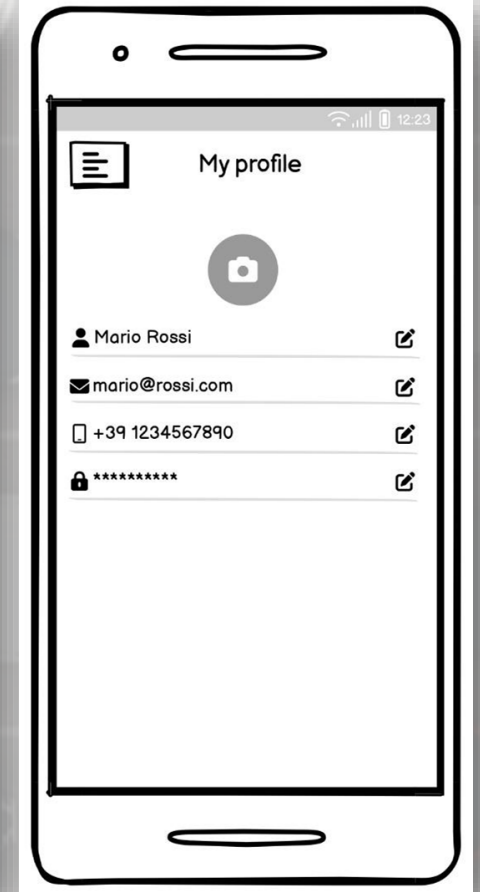
Map, filter, and menù section

Parking page, map of all parking slot

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My parking



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Create a new account
Login and view profile

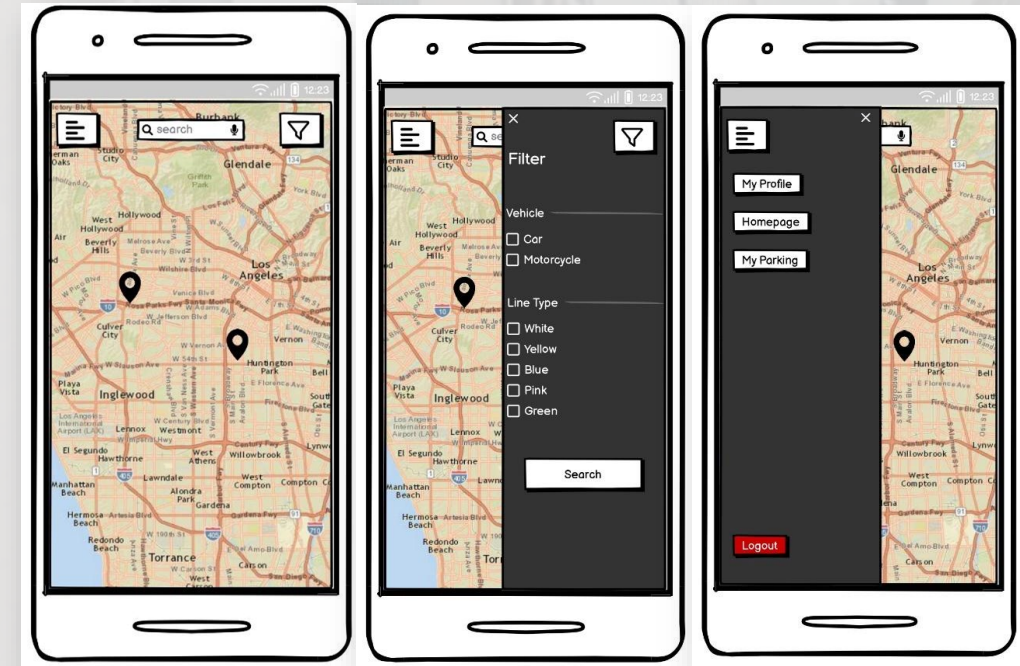
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Create a new account

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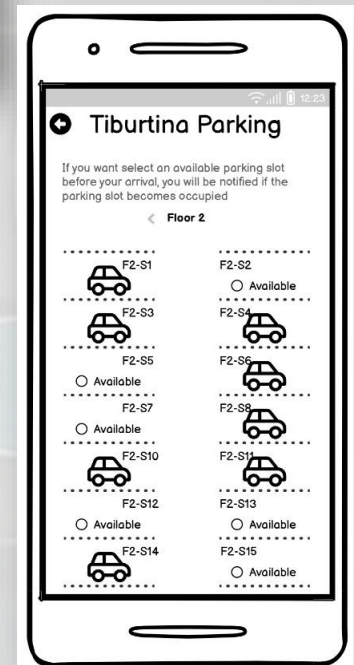
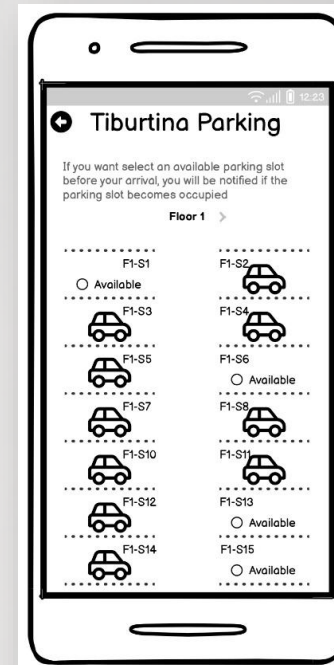
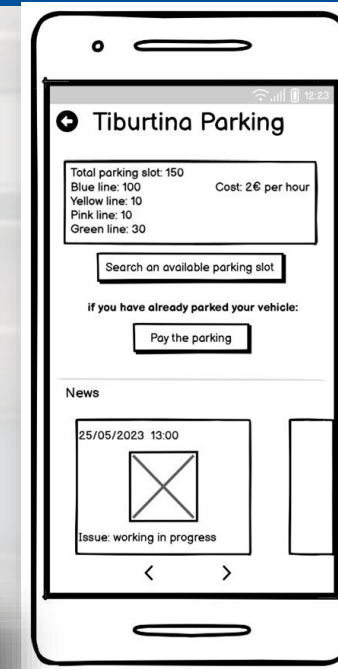
Map, filter, and menù section

Parking page, map of all parking slot

Paying the parking

Extend duration of parking

My parking



Prototype 1



Create a new account
Login and view profile
Map, filter, and menù section
Parking page, map of all parking slot
Paying the parking
Extend duration of parking
My parking

Select payment method

Select an existing payment method

**** * 2540

**** * 1489

Or, Add new card

Owner Name

Card Number

Expiry date: MM YY

CVC

Next

Add parking information

Add slot number

Slot number

Add information about a new vehicle

License plate

Vehicle brand

Vehicle model

Or,

Select a saved vehicle

Add parking duration

hours: 3 minutes: 18

Quick selection duration

1 hr 2 hr 3 hr 4 hr 5 hr

Next

Parking summary

Parking time 4 hr

Parking expiry time 10th June - 17:00

Vehicle details AB234BC
Fiat 500

Parking slot Tiburtina parking
F1-S8 (floor 1, slot 8)

Payment method Credit card
**** * 2540

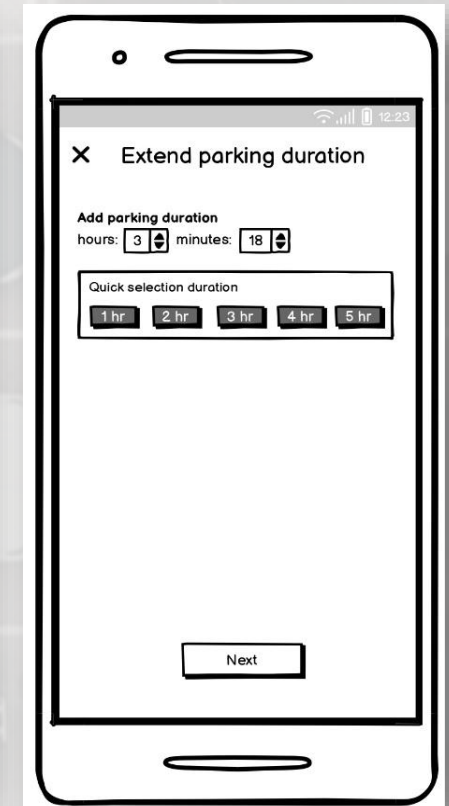
Payment Total €8

Pay

Prototype 1



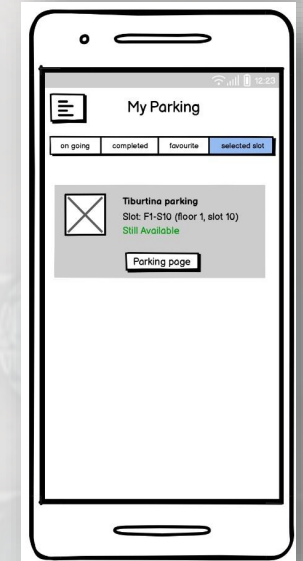
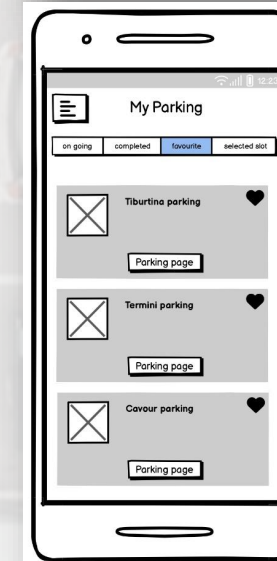
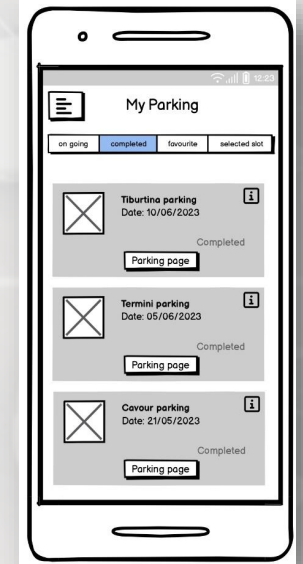
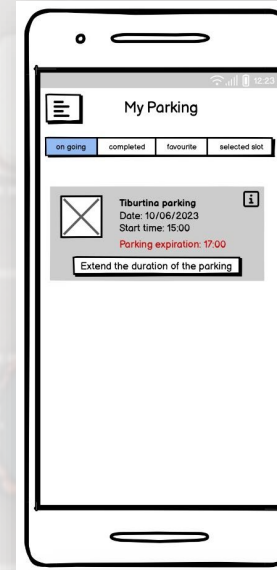
Create a new account
Login and view profile
Map, filter, and menu section
Parking page, map of all parking slot
Paying the parking
Extend duration of parking
My parking



Prototype 1



Create a new account
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My parking





EXPERT BASED EVALUATION



ESPAK

Cognitive Walkthrough



Task 1: *“Pay for a parking with a duration of 3:30 hours, selecting an already saved vehicle, and using your saved credit card.”*

Action 1: Select an already saved vehicle

Response 1: The system shows the selected vehicle.

Action 2: Select the duration of the parking on 3:30 hours.

Response 2: The system shows the selected time duration of the parking.

Action 3: Press the “Next” button to access the next page.

Response 3: The system shows the next page to select the payment method.

A screenshot of a mobile application interface titled "Add parking information". The interface is displayed on a smartphone frame. At the top, there is a status bar with signal strength, Wi-Fi, and battery icons, and the time 12:30. Below the title, there is a section "Add information about a new vehicle" with three input fields: "License plate", "Vehicle brand", and "Vehicle model". Below these fields, there is a section "Or," followed by a dropdown menu labeled "Select a saved vehicle". Below this, there is a section "Add parking duration" with two spinners: "hours" set to 3 and "minutes" set to 18. Below the spinners, there is a section "Quick selection duration" with five buttons: "1 hr", "2 hr", "3 hr", "4 hr", and "5 hr". At the bottom of the form, there is a "Next" button. The background of the slide shows a parking lot with several cars parked.

Cognitive Walkthrough

Action 4: Select your saved credit card.

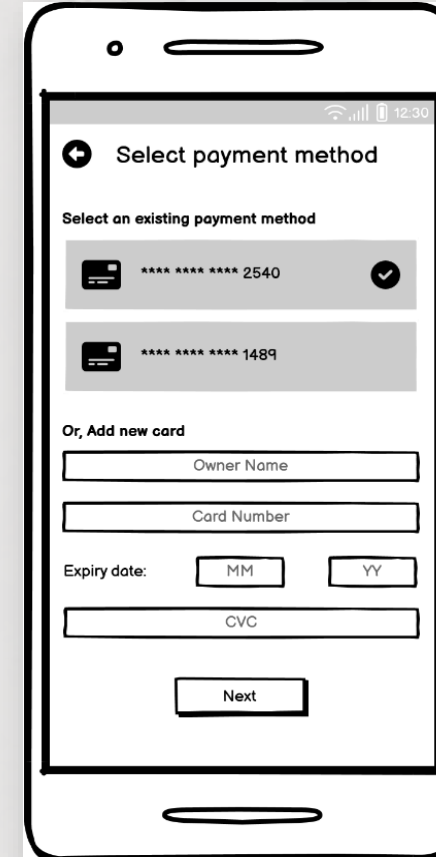
Response 4: The system shows the selected credit card.

Action 5: Press the “Next” button to access the next page.

Response 5: The system shows the next page with a parking summary.

Action 6: Press the “Pay” button to pay the parking.

Response 6: The data is collected by the system and the parking is paid correctly.

A mobile app screen titled "Select payment method". It has a back arrow in the top left. The main heading is "Select an existing payment method". There are two card options: the first is selected with a checkmark and shows "**** * 2540"; the second shows "**** * 1489". Below this is a section "Or, Add new card" with input fields for "Owner Name", "Card Number", "Expiry date" (split into MM and YY), and "CVC". A "Next" button is at the bottom.

Select payment method

Select an existing payment method

**** * 2540 ✓

**** * 1489

Or, Add new card

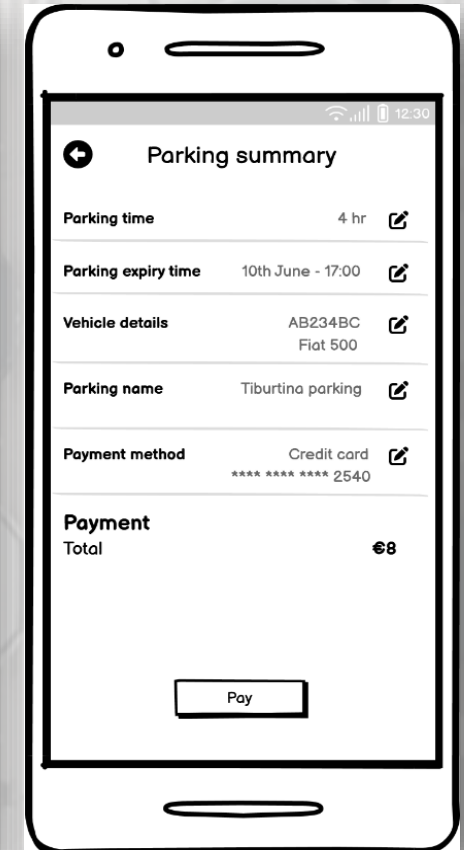
Owner Name

Card Number

Expiry date: MM YY

CVC

Next

A mobile app screen titled "Parking summary". It has a back arrow in the top left. It displays a list of details: "Parking time" (4 hr), "Parking expiry time" (10th June - 17:00), "Vehicle details" (AB234BC, Fiat 500), "Parking name" (Tiburtina parking), and "Payment method" (Credit card, **** * 2540). At the bottom, it shows "Payment Total" as €8 and a "Pay" button.

Parking summary

Parking time 4 hr

Parking expiry time 10th June - 17:00

Vehicle details AB234BC
Fiat 500

Parking name Tiburtina parking

Payment method Credit card
**** * 2540

Payment Total €8

Pay

Cognitive Walkthrough



Task 2: *Pay for a parking with a duration of 2:00 hours, using a new vehicle: 'Volkswagen' car, model 'T Roc', license plate 'FB619ZY', and using a new credit card to pay.*

Action 1: Press text field input and digit the license plate FB619ZY

Response 1: Each digit is displayed as typed, flashing cursor moves to next position.

Action 2: Press text field input and digit the vehicle brand Volkswagen.

Response 2: Each digit is displayed as typed, flashing cursor moves to next position.

Action 3: Press text field input and digit the vehicle model TRoc.

Response 3: Each digit is displayed as typed, flashing cursor moves to next position.

Action 4: Select the duration of the parking on 2:00 hours, using “quick selection duration”.

Response 4: The system shows the selected time duration of the parking.

Action 5: Press the “Next” button to acces the next page.

Response 5: The system shows the next page to select the payment method.

A screenshot of a mobile application interface titled "Add parking information". The interface includes a close button (X) at the top left. Below the title, there is a section "Add information about a new vehicle" with three text input fields: "License plate", "Vehicle brand", and "Vehicle model". Below these fields, there is an "Or," label and a dropdown menu labeled "Select a saved vehicle". Further down, there is a section "Add parking duration" with two spinners for "hours" (set to 3) and "minutes" (set to 18). Below this, there is a "Quick selection duration" section with five buttons: "1 hr", "2 hr", "3 hr", "4 hr", and "5 hr". At the bottom of the form, there is a "Next" button. The background of the app shows a grid of parking spaces with cars.

Cognitive Walkthrough

Action 6: Press text field input and digit the Owner Name “Lorenzo Russo”.

Response 6: Each digit is displayed as typed, flashing cursor moves to next position.

Action 7: Press text field input and digit the Card Number “12345678”.

Response 7: Each digit is displayed as typed, flashing cursor moves to next position.

Action 8: Press text field input and digit the Expiry date (month and year), “10/2028”.

Response 8: Each digit is displayed as typed, flashing cursor moves to next position.

Action 9: Press text field input and digit the CVC code “123”.

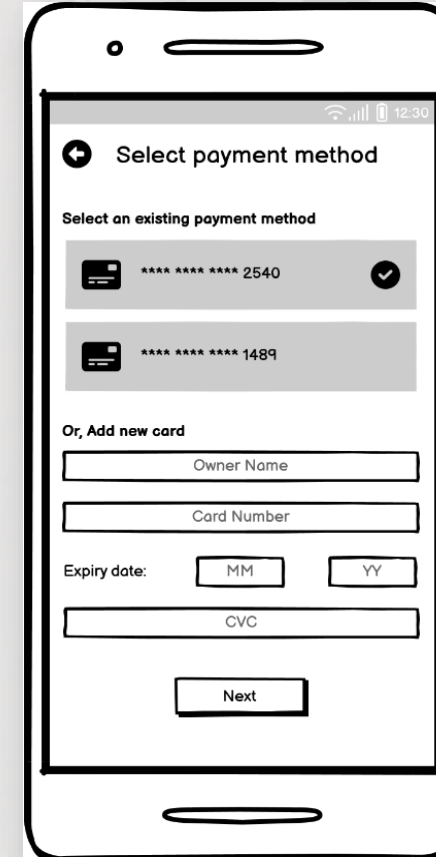
Response 9: Each digit is displayed as typed, flashing cursor moves to next position.

Action 10: Press the “Next” button to access the next page.

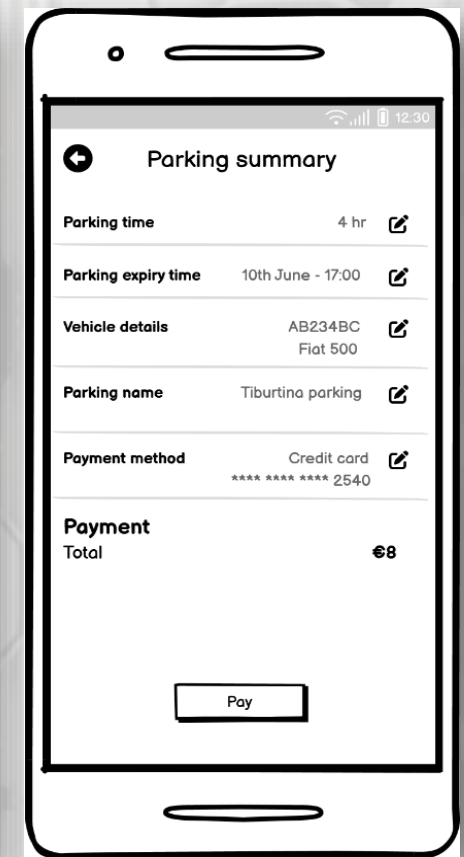
Response 10: The system shows the next page with a parking summary.

Action 11: Press the “Pay” button to pay the parking.

Response 11: The data is collected by the system and the parking is paid correctly.



The screenshot shows a mobile app interface for selecting a payment method. At the top, there's a back arrow and the title "Select payment method". Below this, a section titled "Select an existing payment method" contains two card options. The first card is selected, indicated by a checkmark, and shows a masked card number ending in 2540. The second card is unselected and shows a masked card number ending in 1489. Below these options, there's a section titled "Or, Add new card" which contains four input fields: "Owner Name", "Card Number", "Expiry date:" (with MM and YY sub-fields), and "CVC". At the bottom of this section is a "Next" button.



The screenshot shows a mobile app interface for a parking summary. At the top, there's a back arrow and the title "Parking summary". Below this, there are several rows of information, each with a right arrow icon for editing: "Parking time" (4 hr), "Parking expiry time" (10th June - 17:00), "Vehicle details" (AB234BC Fiat 500), "Parking name" (Tiburtina parking), and "Payment method" (Credit card **** * 2540). At the bottom, there's a "Payment" section showing "Total" as €8, and a "Pay" button.

Heuristic Evaluation

Heuristic Evaluation - ESPark, 15/06/23

Page	Heuristic violated	Severity (1-5)	Description/comment
Login	Help users recognize, diagnose, and recover from errors	4	Include a "Forgot password" link
Tiburtina Parking page	Match between system and real world, Help and documentation	4	It is not clear the description of the parking area - is it a public parking area on the street, or a private facility as the mall example in the introduction? Also, it is not clear what the line classification stands for. I assume it refers to the classification of street parking spots in Rome, but it may not be shared between all municipalities (hence between all users). In this case, it would suggest the parking area is public. On the other hand, the news section seems more a private thing. Having said so, the parking type is not reported in the parking slot detail, so what is its purpose?
Tiburtina Parking page	Visibility of system status, Flexibility and efficiency of use	4	Why ask the user to "search for a parking spot", if the system already knows the situation? Which is the label if no parking spot is available?
Homepage (map)	Match between system and real world, Consistency and standards	3	Indicating something with a pin on the map usually indicates a single activity (e.g. a single and private parking facility). If it is <u>public</u> I would suggest indicating it with an area rather than the single pin.

			(Menu>My Parking>On going), but also in the homepage could be useful.
General comment 1	Match between the system and the real world	4	It may seem that you are allowing to book and pay for specific parking spots (e.g. the chance of selecting a parking spot in the map, but also in the payment form, there is the section for the "parking slot" selected). However, reading more carefully (e.g. the note "...you will be notified if the parking slot becomes occupied") you understand that the app is just providing info about the availability status of the spots. I would suggest an overall clarification about this, as it is the main feature of the app.

Severity legend:

- 0= I don't agree that this is a usability problem at all
- 1= Cosmetic problem only
- 2= Minor usability problem
- 3= Major usability problem
- 4= Usability catastrophe

Based on Jakob Nielsen's 10 Usability Heuristics:

1. Visibility of system status
2. Match between the system and the real world
3. User control and freedom
4. Consistency and standards
5. Error prevention
6. Recognition rather than recall
7. Flexibility and efficiency of use
8. Aesthetic and minimalist design
9. Help users recognize, diagnose and recover from errors
10. Help and documentation



PROTOTYPE 2

Prototype 2



Create a new account

Login and view profile

Map, filter, and menù section

Parking page, map of all parking slot

Paying the parking

Extend duration of parking

My parking

16:39

✕

Create your Account

Email

Password

Sign up

or continue with

f G

Already have an account? Sign in

16:40

← Fill Your Profile

Name

Surname

Date of Birth

Phone Number

SIGN UP

Prototype 2



Create a new account

Login and view profile

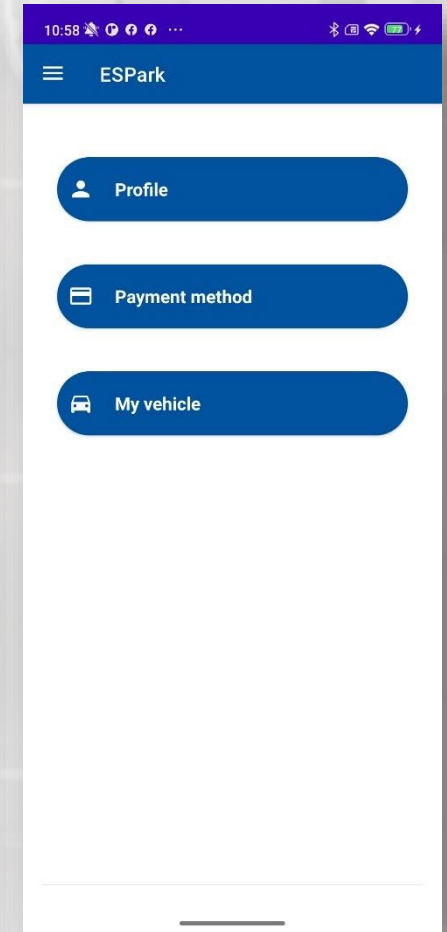
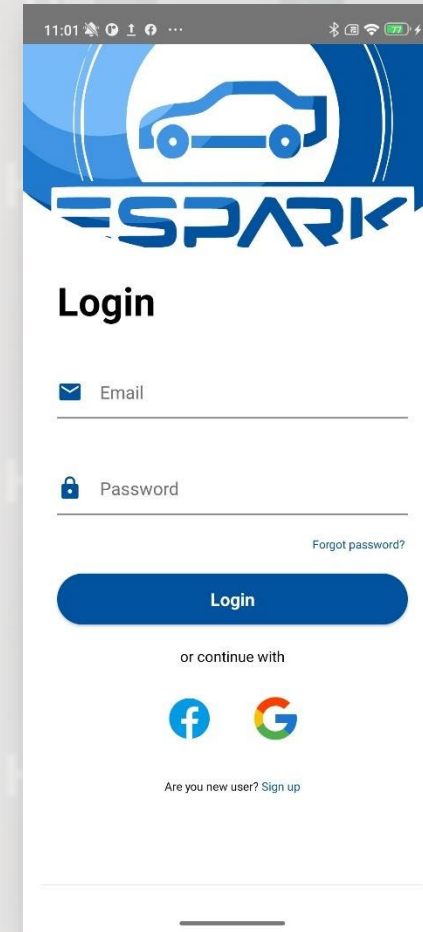
Map, filter, and menu section

Parking page, map of all parking slot

Paying the parking

Extend duration of parking

My parking



Expert tell us to include a “forgot password” link

Prototype 2



Create a new account
Login and view profile

Map, filter, and menù section

Parking page, map of all parking slot

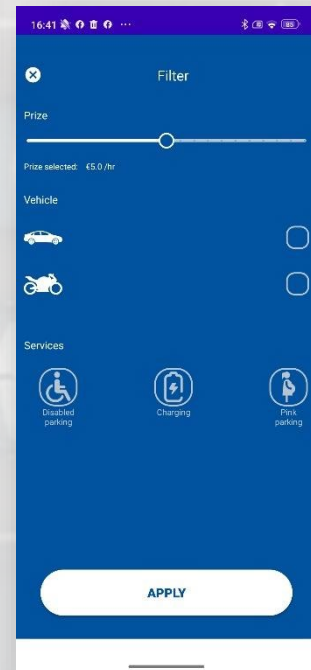
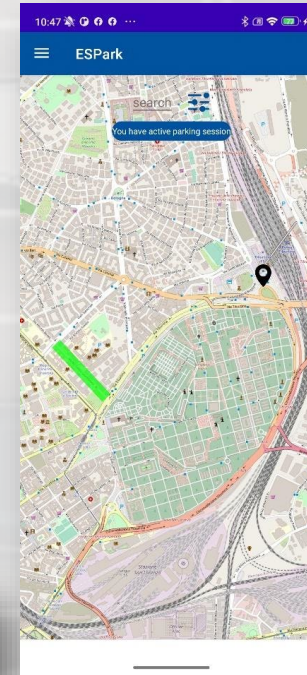
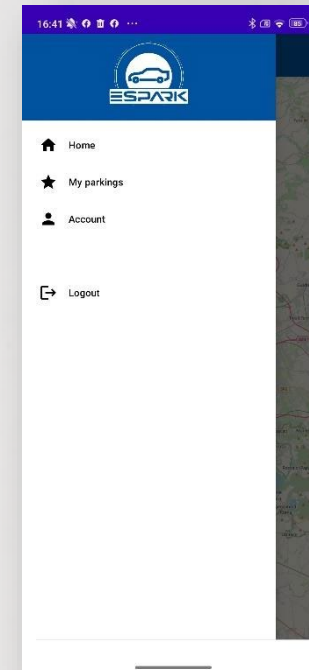
Paying the parking

Extend duration of parking

My parking

Expert tell us that if the parking is a single private parking we indicate it with a pin while if it is a public parking we indicate it with a colored area

Expert tell us to add a feedback about having paid a parking spot



Prototype 2



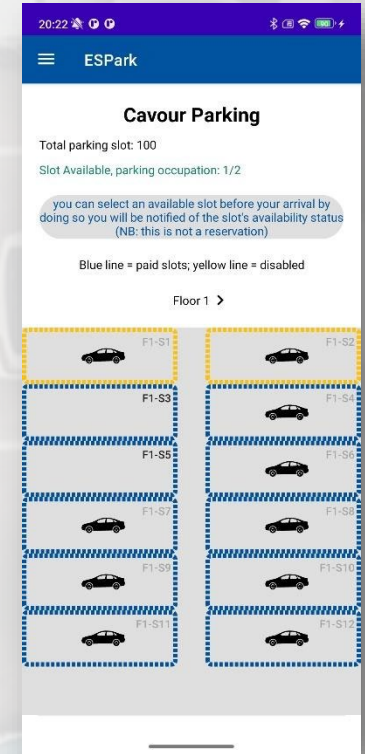
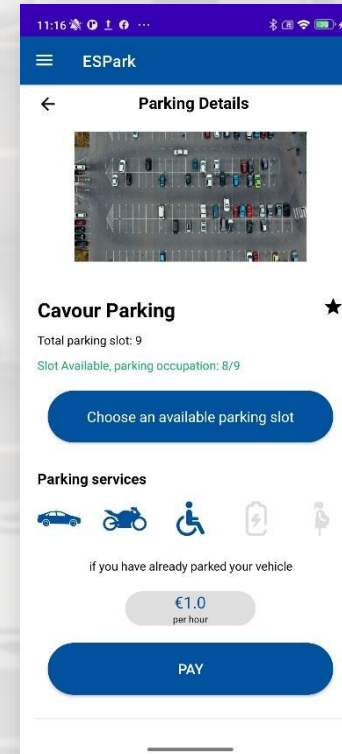
Create a new account
Login and view profile
Map, filter, and menù section

Parking page, map of all parking slot

Paying the parking

Extend duration of parking

My parking



Expert tell us to insert an image that describes the type of parking slot and, as parking services, we have inserted the icon to classify the type of parking.

Prototype 2



Create a new account
Login and view profile
Map, filter, and menù section
Parking page, map of all parking slot
Paying the parking
Extend duration of parking
My parking

The expert told us to eliminate the possibility of insert the slot number, now the user pays for parking using the license plate directly.

10:55 10:57 ...

✕ Add parking information

Parking duration

1 hr

Start Hour 10:55 End Hour 11:55 Total €1.0

Select your Vehicle

☐ Model: Nissan Micra, License plate:AB123CD

☒ Model: Audi A5, License plate:DF564HJ

☐ Model: Fiat Panda, License plate:FI453SD

☐ Model: Audi Q3, License plate:CF574KL

Or add new vehicle +

NEXT

10:57 10:57 ...

✕ Add payment information

Choose your payment method

☒ Type: Mastercard, Number: 123456789

☐ Type: PayPal, Number: 654398761234

Or add new card +

NEXT

10:57 10:57 ...

✕ Parking summary

Parking name Cavour Parking

Start date 5-9 10:57

Expiry date 5-9 11:57

Duration 1 hr

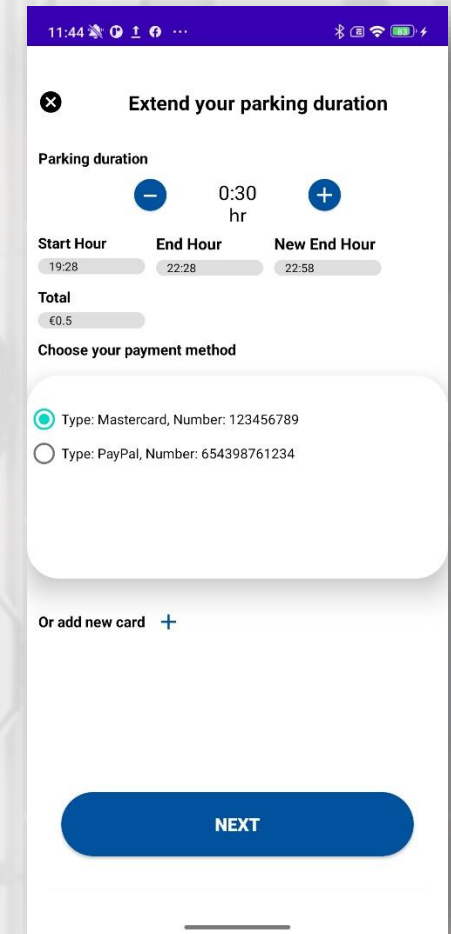
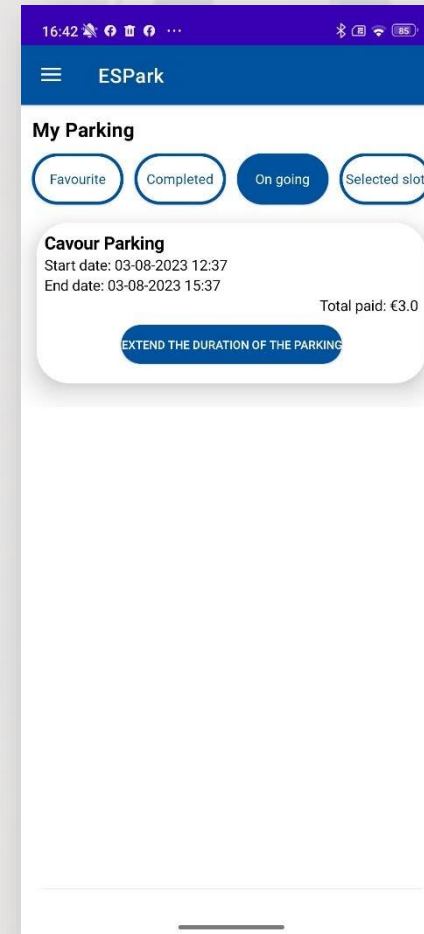
Vehicle Audi A5 (DF564HJ)

Payment method 123456789, Mastercard

Total €1.0

PAY

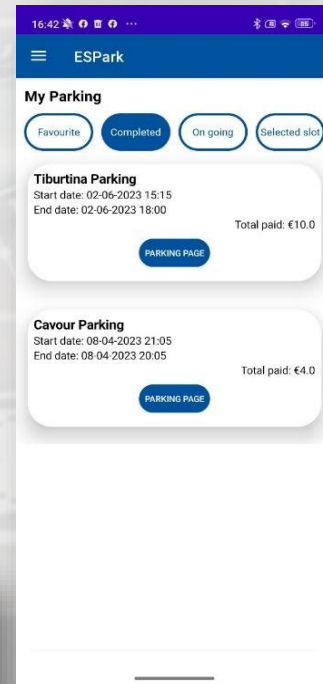
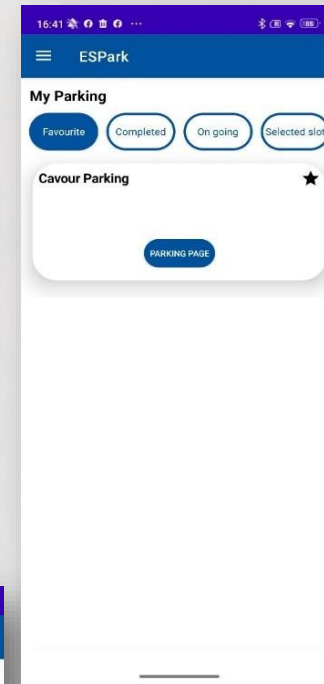
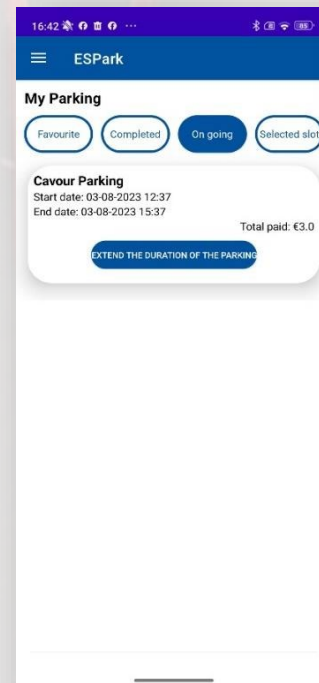
Create a new account
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Prototype 2



Create a new account
Login and view profile
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Paying the parking
Extend duration of parking
My parking





**USER BASED
EVALUATION:
THINK ALOUD**

10 participants who all fell within the user profile of this application.

Tasks executed by users:

1. Look for a parking using the filters and then look for an available parking slot in the chosen parking
2. Pay the parking
3. Extend parking duration

The user comments all during the execution of the task

After the execution of experiment:

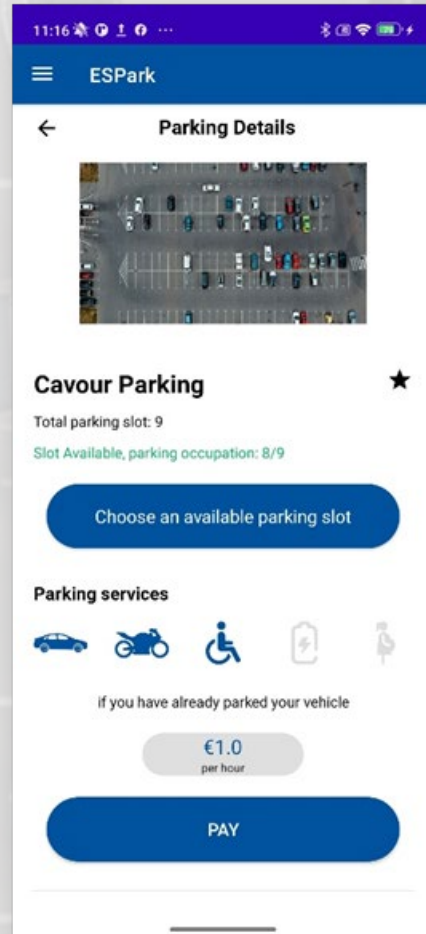
Tasks 2 and 3 did not have any particular problems being carried out by the users.

For the task 1 we identified 2 issues.

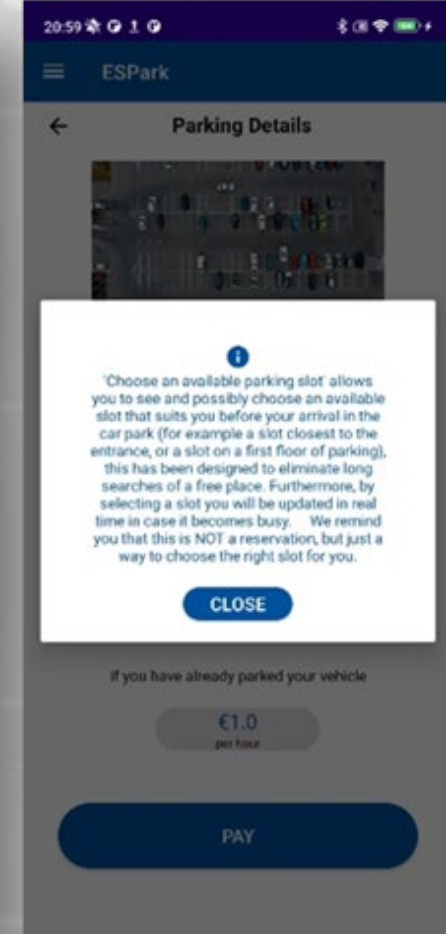
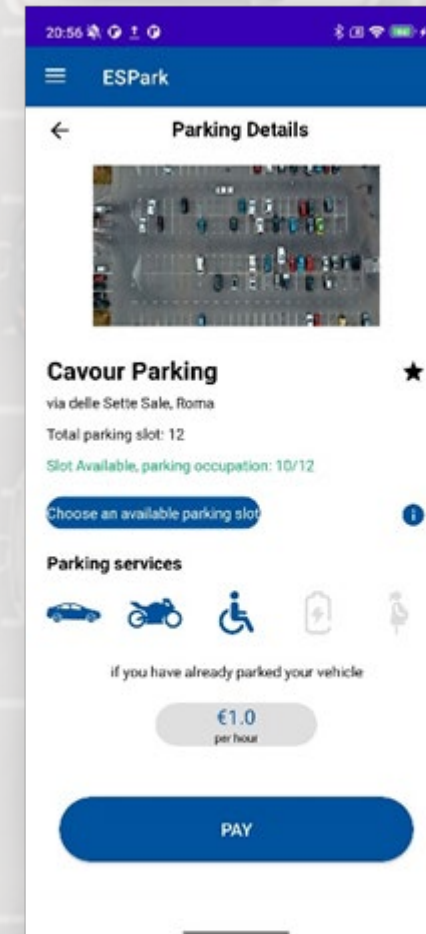
Think Aloud

Task: Look for a parking using the filters and then look for an available parking slot in the chosen parking.

Issue 1: some users had difficulty understanding the task “Choose an available parking slot”



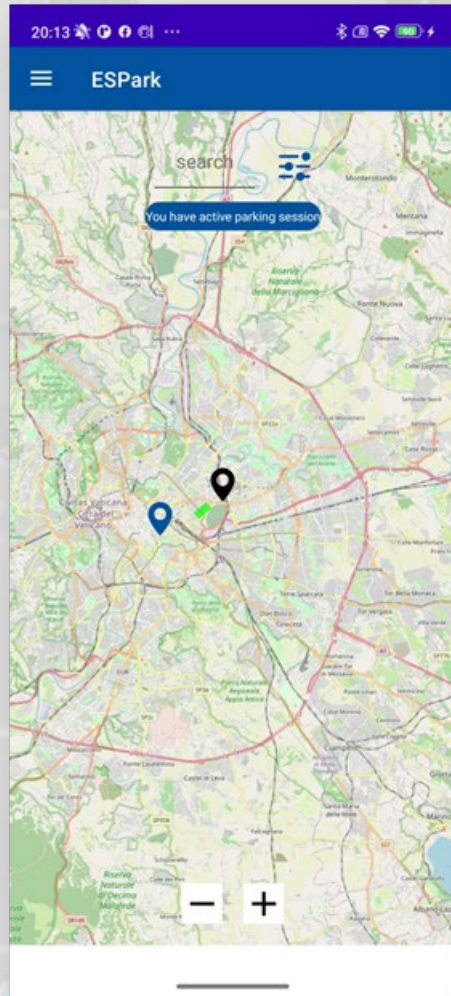
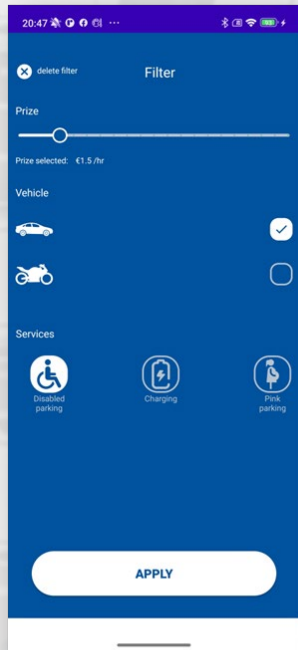
Solution 1: add info button to obtain more information on the service offered



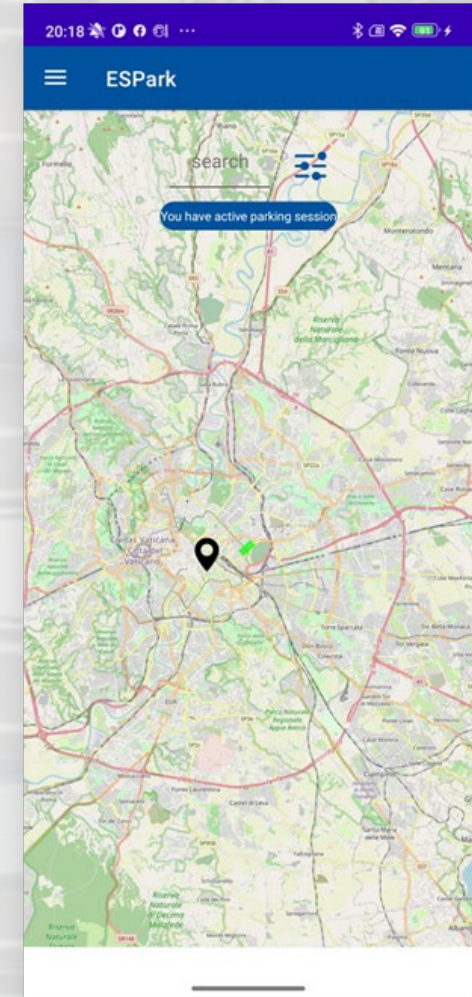
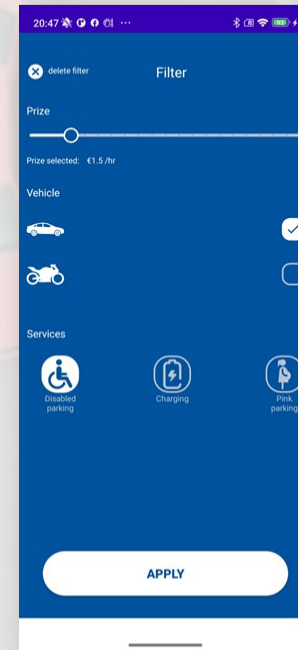
Think Aloud

Task: Look for a parking using the filters and then look for an available parking slot in the chosen parking.

Issue 2: difficulty in identifying car parks that respect certain filters (which were highlighted) after having inserted the filters



Solution 2: display on the map only the car parks that comply with the entered filters.





USER BASED EVALUATION: CONTROLLED EXPERIMENT



Controlled Experiment - experiment 1

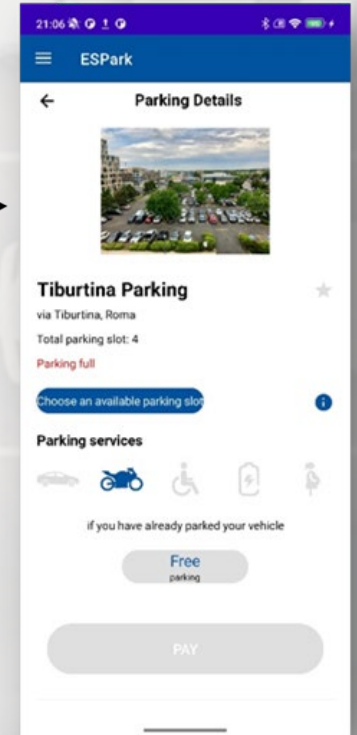
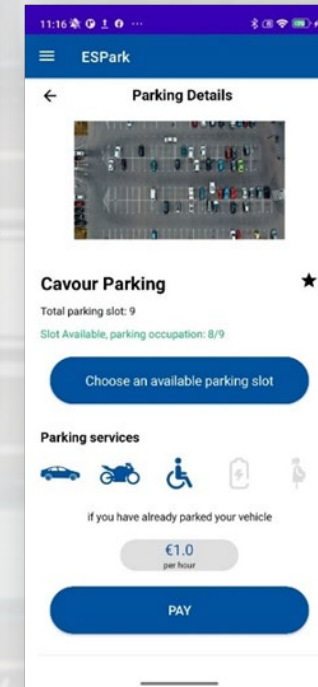
Task 1: Look for a parking with available slots using the map.



Interface 1



Interface 2



Controlled Experiment - experiment 1

- 10 participants
- Independent variables: interface 1 and interface 2
- Dependent variables: times taken by each user to complete the task
- Null hypothesis: both interfaces are equal
- Our hypothesis: interface 1 is better than interface 2
- Method: between-group

ANOVA analysis and results

Interface 1	Interface 2
7,5	25,1
16,2	30,2
10,8	16,8
8,3	20,3
6,5	11,7

Analisi varianza: ad un fattore

RIEPILOGO

Gruppi	Conteggio	Somma	Media	Varianza
Colonna 1	5	49,3	9,86	15,093
Colonna 2	5	104,1	20,82	51,477

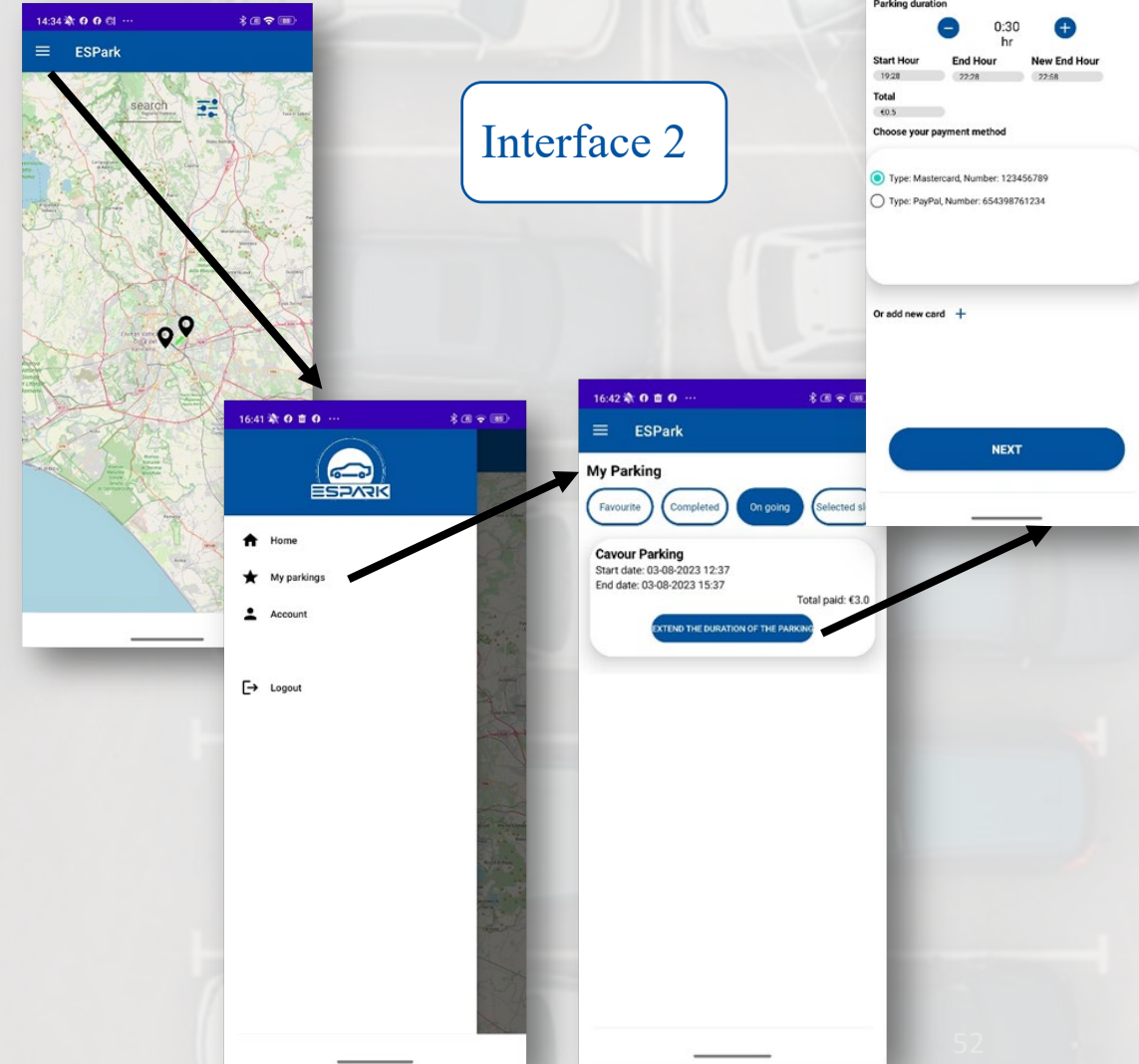
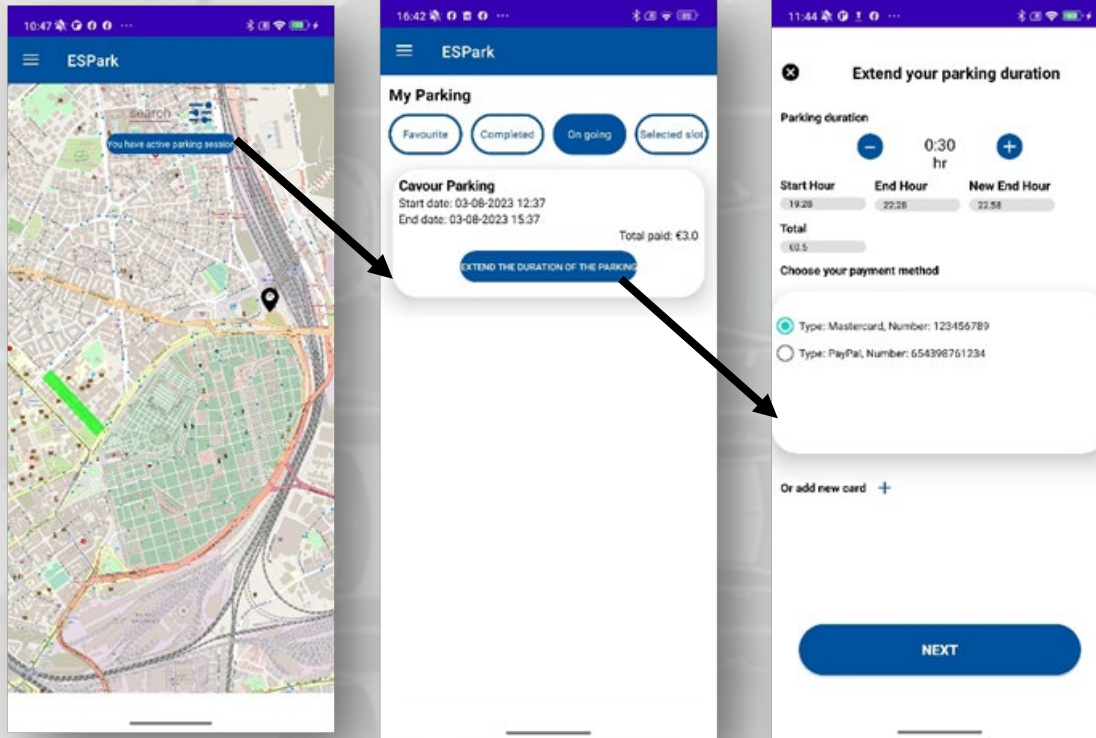
ANALISI VARIANZA

Origine della variazione	SQ	gdl	MQ	F	Valore di significatività	F crit
Tra gruppi	300,304	1	300,304	9,022202	0,016975743	5,317655
In gruppi	266,28	8	33,285			
Totale	566,584	9				

Controlled Experiment - experiment 2



Task 2: Extend the duration of parking



Controlled Experiment - experiment 2

- 10 participants
- Independent variables: interface 1 and interface 2
- Dependent variables: times taken by each user to complete the task
- Null hypothesis: both interfaces are equal
- Our hypothesis: interface 1 is better than interface 2
- Method: within-group

ANOVA analysis and results

Interface 1	Interface 2
8,7	28,7
9,5	35,6
6,5	24,2
10,7	41,3
7,8	23,4
8,6	18,5
9,1	15,7
11,8	25,8
10,3	22,9
13,9	27,2

Analisi varianza: ad un fattore

RIEPILOGO

Gruppi	Conteggio	Somma	Media	Varianza
Colonna 1	10	96,9	9,69	4,4298889
Colonna 2	10	263,3	26,33	57,320111

ANALISI VARIANZA

Origine della variazione	SQ	gdl	MQ	F	Valore di significatività	F crit
Tra gruppi	1384,448	1	1384,448	44,840421	2,79579E-06	4,413873
In gruppi	555,75	18	30,875			
Totale	1940,198	19				

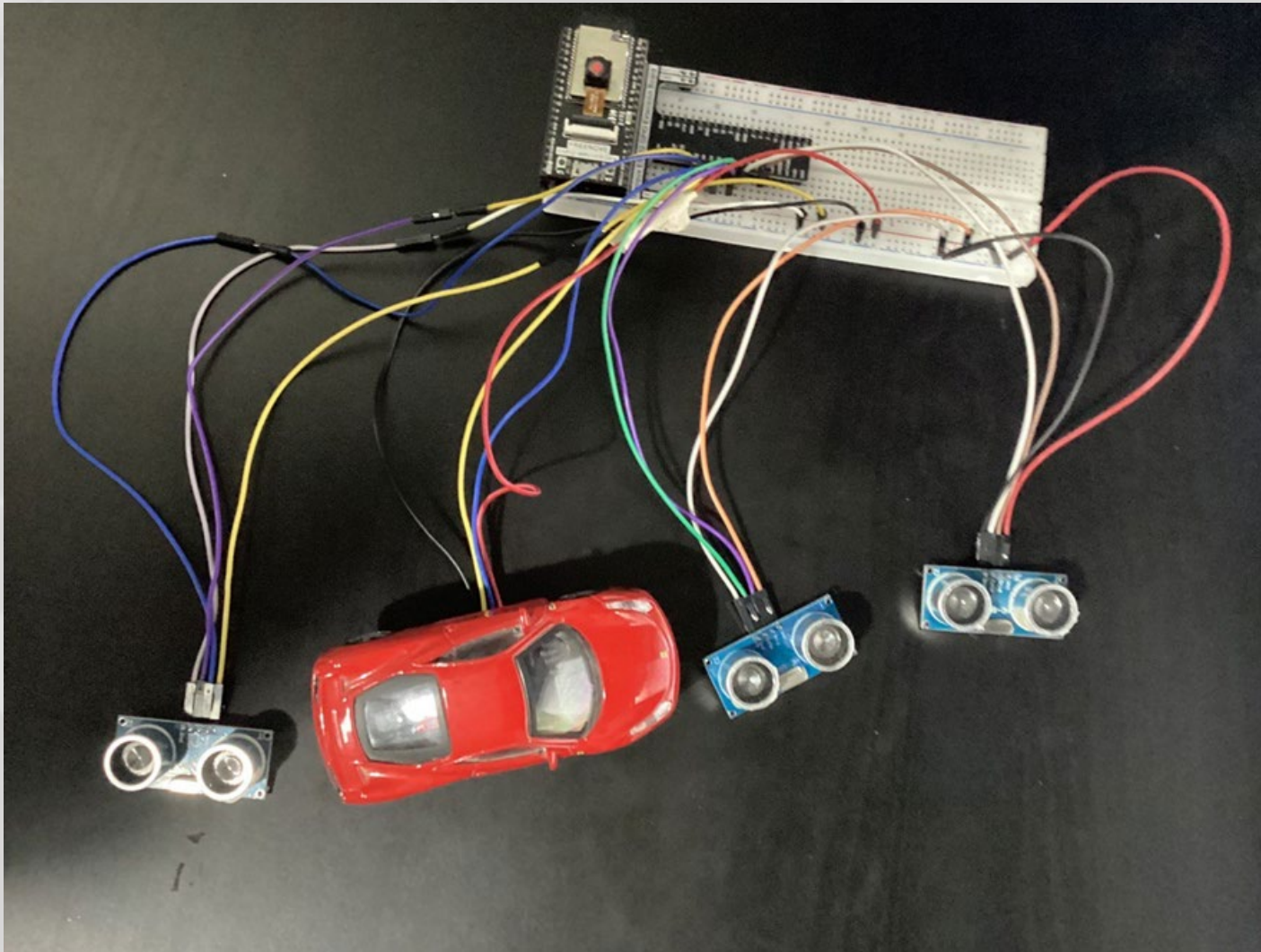


IMPLEMENTATION AND FINAL PRODUCT



ESPAK

Hardware part



We use the ESP32 microcontroller to simulate a real car park where there are 4 sensors representing the car slots, in such away as to simulate the availability of the slot in real time.

We use Arduino IDE to program the microcontroller in C.

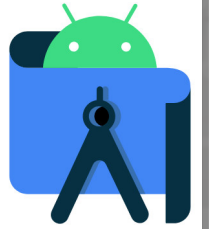


Arduino IDE

We created the app with Android Studio.

We use Firebase to create a Realtime Database and with this it was possible to implement real-time visualization of available slots from the microcontroller to the all user that use the app ESPark.

android
studio



Firebase

THANK YOU FOR THE ATTENTION

We now show a demo of the real system